



Abhigyan Gurukul

Q Akashar Pavillion, Yogi Nagar, Near Priya Cinema, Vadodara, Gujarat

+919472600836, +919829491219

admin@abhigyangurukul.com

abhigyangurukul.com

PHYSICS - 05

Date : _____

Name : _____

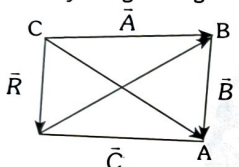
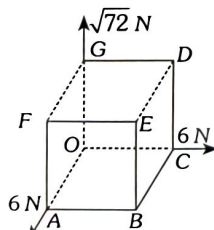
Roll No. : _____

TOPIC: Fundamental of Vectors, Addition, Subtraction and Multiplication of Vectors

- Which of the following is vector quantity?
(a) Coefficient of friction (b) Temperature
(c) Charge (d) Impulse
- If θ is the angle between two vectors, then the resultant vector is maximum, when value of θ is:
(a) 0° (b) 90°
(c) 180° (d) same in all cases
- In a clock-wise system,
(a) $\hat{j} \times \hat{k} = \hat{i}$ (b) $\hat{i} \times \hat{k} = 0$
(c) $\hat{j} \times \hat{j} = \hat{i}$ (d) $\hat{k} \times \hat{i} = \hat{j}$
- Which of the following statements is/are incorrect?
(i) A scalar quantity is the one that is conserved in a process.
(ii) A scalar quantity is the one that can never take negative values.
(iii) A scalar quantity has the same value for observers with different orientations of the axis.
(a) (i) and (iii) (b) (ii) only
(c) (ii) and (iii) (d) (i) and (ii)
- How many minimum numbers of vectors in different plane can be added to give zero resultant?
(a) 2 (b) 3
(c) 4 (d) 5
- The unit vectors along the three co-ordinate axes are related as:
(a) $\hat{i} > \hat{j} > \hat{k} > 1$ (b) $\hat{i} = \hat{j} = \hat{k} = 0$
(c) $\hat{i} = -\hat{j} = \hat{k} = 1$ (d) $\hat{i} = \hat{j} = \hat{k} = 1$
- Identify the pairs of vectors which are parallel to each other?
(a) $\vec{A} = \hat{i} - 2\hat{j}; \vec{B} = \hat{i} - 5\hat{j}$ (b) $\vec{A} = \hat{i} - 10\hat{j}; \vec{B} = 2\hat{i} - 5\hat{j}$
(c) $\vec{A} = \hat{i} - 5\hat{j}; \vec{B} = \hat{i} - 10\hat{j}$ (d) $\vec{A} = \hat{i} - 5\hat{j}; \vec{B} = 2\hat{i} - 10\hat{j}$
- The angle between the direction of $\hat{i}$ and $|\hat{i} + \hat{j}|$ is:
(a) 90° (b) 0°
(c) 45° (d) 180°
- Two vectors $\vec{A}$ and $\vec{B}$ inclined at an angle θ have a resultant $\vec{C}$ which makes an angle β with $\vec{A}$. If the directions of $\vec{A}$ and $\vec{B}$ are interchanged, the resultant will have the same
(a) direction.
(b) magnitude.
(c) direction as well as magnitude.
(d) None of these
- The vector $\vec{a} = m\hat{i} + 2\hat{j} + n\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ that bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of m and n ?
(a) $m = 2, n = 2$ (b) $m = 1, n = 2$
(c) $m = 2, n = 1$ (d) $m = 1, n = 1$

RESPONSE
GRID

- | | | | | |
|--------------------|--------------------|--------------------|--------------------|---------------------|
| 1. (a) (b) (c) (d) | 2. (a) (b) (c) (d) | 3. (a) (b) (c) (d) | 4. (a) (b) (c) (d) | 5. (a) (b) (c) (d) |
| 6. (a) (b) (c) (d) | 7. (a) (b) (c) (d) | 8. (a) (b) (c) (d) | 9. (a) (b) (c) (d) | 10. (a) (b) (c) (d) |

11. Let $\vec{A} = \hat{i}A \cos \theta + \hat{j}A \sin \theta$ be any vector. Another vector $\vec{B}$, which is normal to $\vec{A}$ can be expressed as
- (a) $\hat{i}B \cos \theta - \hat{j}B \sin \theta$ (b) $\hat{i}B \cos \theta + \hat{j}B \sin \theta$
 (c) $\hat{i}B \sin \theta - \hat{j}B \cos \theta$ (d) $\hat{i}B \sin \theta + \hat{j}B \cos \theta$
12. Given $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$, which of the following statements is not correct?
- (a) $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ must each be a null vector.
 (b) The magnitude of $(\vec{A} + \vec{C})$ equals the magnitude of $(\vec{B} + \vec{D})$.
 (c) The magnitude of $\vec{A}$ can never be greater than the sum of the magnitudes of $\vec{B}$, $\vec{C}$, and $\vec{D}$
 (d) $(\vec{B} + \vec{C})$ must lie in the plane of $\vec{A}$ and $\vec{D}$ if $\vec{A}$ and $\vec{D}$ are not collinear and in the line of $\vec{A}$ and $\vec{D}$, if they are collinear.
13. Which law is governed by the given figure?
- 
- (a) Associative law of vector addition
 (b) Commutative law of vector addition
 (c) Associative law of vector multiplication
 (d) Commutative law of vector multiplication
14. Three forces of magnitudes 6 N, 6 N, and $\sqrt{72}$ N act at corner of cube along three sides as shown in figure. Resultant of these forces is:
- 
- (a) 12 N angle OB
 (b) 18 N along OA
 (c) 18 N along OC
 (d) 12 N along OE
15. The magnitude of the x-component of vector $\vec{A}$ is 3 and the magnitude of vector $\vec{A}$ is 5. What is the magnitude of the y-component of vector $\vec{A}$?
- (a) 3 (b) 4
 (c) 5 (d) 8
16. Two forces are such that the sum of their magnitudes is 18 N and their resultant is 12 N which is perpendicular to the smaller force. Then the magnitudes of the forces are:
- (a) 12 N, 6 N (b) 13 N, 5 N
 (c) 10 N, 8 N (d) 16 N, 2 N
17. Which one is a vector quantity?
- (a) Time (b) Temperature
 (c) Magnetic flux (d) Magnetic field intensity
18. What is the dot product of two vectors of magnitudes 3 and 5, if angle between them is 60° ?
- (a) 5.2 (b) 7.5
 (c) 8.4 (d) 8.6
19. The forces, which meet at one point but their lines of action do not lie in one plane, are called
- (a) non-coplanar non-concurrent forces
 (b) non-coplanar concurrent forces
 (c) coplanar concurrent forces
 (d) coplanar non-concurrent forces
20. A vector $\vec{A}$ points vertically upward and $\vec{B}$ points towards north. The vector product $\vec{A} \times \vec{B}$ is
- (a) along west (b) along east
 (c) zero (d) vertically downward
21. A force $(3\hat{i} + 4\hat{j})$ newton acts on a body and displaces it by $(3\hat{i} + 4\hat{j})$ metre. The work done by the force is
- (a) 5 J (b) 25 J
 (c) 10 J (d) 30 J
22. The torque of force $F = (2\hat{i} + 3\hat{j} + 4\hat{k})$ newton acting at the point $r = (3\hat{i} + 2\hat{j} + 3\hat{k})$ metre about origin is (in N-m)
- (a) $6\hat{i} + 6\hat{j} + 12\hat{k}$ (b) $17\hat{i} + 6\hat{j} + 13\hat{k}$
 (c) $-6\hat{i} + 6\hat{j} - 12\hat{k}$ (d) $-17\hat{i} + 6\hat{j} - 13\hat{k}$
23. Maximum and minimum values of the resultant of two forces acting at a point are 7 N and 3 N respectively. The smaller force will be equal to
- (a) 5 N (b) 4 N
 (c) 2 N (d) 1 N

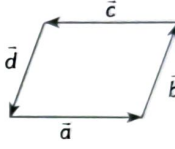
RESPONSE
GRID

11. (a) (b) (c) (d) 12. (a) (b) (c) (d) 13. (a) (b) (c) (d) 14. (a) (b) (c) (d) 15. (a) (b) (c) (d)
 16. (a) (b) (c) (d) 17. (a) (b) (c) (d) 18. (a) (b) (c) (c) 19. (a) (b) (c) (d) 20. (a) (b) (c) (d)
 21. (a) (b) (c) (d) 22. (a) (b) (c) (d) 23. (a) (b) (c) (d)

24. If the vectors $\vec{P} = a\hat{i} + a\hat{j} + 3\hat{k}$ and $\vec{Q} = a\hat{i} - 2\hat{j} - \hat{k}$ are perpendicular to each other, then the positive value of a is
 (a) zero (b) 1
 (c) 2 (d) 3
25. The (x, y, z) co-ordinates of two points A and B are given respectively as $(0, 3, -1)$ and $(-2, 6, 4)$. The displacement vector from A to B is given by
 (a) $-2\hat{i} + 6\hat{j} + 4\hat{k}$ (b) $-2\hat{i} + 3\hat{j} + 3\hat{k}$
 (c) $-2\hat{i} + 3\hat{j} + 5\hat{k}$ (d) $2\hat{i} - 3\hat{j} - 5\hat{k}$
26. A vector is not changed if
 (a) it is rotated through an arbitrary angle
 (b) it is multiplied by an arbitrary scalar
 (c) it is cross multiplied by a unit vector
 (d) it is displaced parallel to itself
27. The angles which the vector $\vec{A} = 3\hat{i} + 6\hat{j} + 2\hat{k}$ makes with the co-ordinate axes are
 (a) $\cos^{-1} \frac{3}{7}, \cos^{-1} \frac{6}{7}$ and $\cos^{-1} \frac{2}{7}$
 (b) $\cos^{-1} \frac{4}{7}, \cos^{-1} \frac{5}{7}$ and $\cos^{-1} \frac{3}{7}$
 (c) $\cos^{-1} \frac{3}{7}, \cos^{-1} \frac{4}{7}$ and $\cos^{-1} \frac{1}{7}$
 (d) None of these
28. Unit vector parallel to the resultant of vectors $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 8\hat{i} + 8\hat{j}$ will be
 (a) $\frac{24\hat{i} + 5\hat{j}}{13}$ (b) $\frac{12\hat{i} + 5\hat{j}}{13}$
 (c) $\frac{6\hat{i} + 5\hat{j}}{13}$ (d) None of these
29. The component of vector $\vec{A} = 2\hat{i} + 3\hat{j}$ along vector $\hat{i} + \hat{j}$
 (a) $\frac{5}{\sqrt{2}}$ (b) $10\sqrt{2}$
 (c) $5\sqrt{2}$ (d) 5
30. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A} + \vec{B} = \vec{C}$ and $A^2 + B^2 = C^2$. If θ is the angle between positive direction of $\vec{A}$ and $\vec{B}$, then the correct statement is
 (a) $\theta = \pi$ (b) $\theta = 2\pi/3$
 (c) $\theta = 0$ (d) $\theta = \pi/2$
31. If $|\vec{A} \times \vec{B}| = \sqrt{2}\vec{A} \times \vec{B}$, then the value of $|\vec{A} + \vec{B}|$ is
 (a) $(A^2 + B^2 + AB)^{1/2}$ (b) $(A^2 + B^2 + \frac{AB}{\sqrt{3}})^{1/2}$
 (c) $(A + B)$ (d) $(A^2 + B^2 + \sqrt{3}AB)^{1/2}$
32. If the angle between the vectors $\vec{A}$ and $\vec{B}$ is θ , the value of the product $(\vec{B} \times \vec{A}) \cdot \vec{A}$ is equal to
 (a) $BA^2 \cos \theta$ (b) $BA^2 \sin \theta$
 (c) $BA^2 \sin \theta \cos \theta$ (d) zero
33. Given that $\vec{P} + \vec{Q} + \vec{R} = 0$. Two out of the three vectors are equal in magnitude. The magnitude of the third vector is $\sqrt{2}$ times that of the other two. Which of the following can be the angles between these vectors?
 (a) $90^\circ, 135^\circ, 135^\circ$ (b) $45^\circ, 45^\circ, 90^\circ$
 (c) $30^\circ, 60^\circ, 90^\circ$ (d) $45^\circ, 90^\circ, 135^\circ$
34. The value of n so that vectors $2\hat{i} + 3\hat{j} - 2\hat{k}, 5\hat{i} + n\hat{j} + \hat{k}$ and $-\hat{i} + 2\hat{j} + 3\hat{k}$ may be coplanar, will be
 (a) 18 (b) 28
 (c) 9 (d) 36
35. The resultant of two forces $3P$ and $2P$ is R . If the first force is doubled then the resultant is also doubled. The angle between the two forces is
 (a) 60° (b) 120°
 (c) 30° (d) 135°
36. The resultant of two forces, one double the other in magnitude, is perpendicular to the smaller of the two forces. The angle between the two forces is
 (a) 120° (b) 60°
 (c) 90° (d) 150°
37. Let the angle between two non-zero vectors $\vec{A}$ and $\vec{B}$ be 120° and its resultant be $\vec{C}$.
 (a) C must be equal to $|A - B|$
 (b) C must be less than $|A - B|$
 (c) C must be greater than $|A - B|$
 (d) C may be equal to $|A - B|$

RESPONSE
GRID

24. (a) (b) (c) (d) 25. (a) (b) (c) (d) 26. (a) (b) (c) (d) 27. (a) (b) (c) (d) 28. (a) (b) (c) (d)
 29. (a) (b) (c) (d) 30. (a) (b) (c) (d) 31. (a) (b) (c) (d) 32. (a) (b) (c) (d) 33. (a) (b) (c) (d)
 34. (a) (b) (c) (d) 35. (a) (b) (c) (d) 36. (a) (b) (c) (d) 37. (a) (b) (c) (d)

38. OP is a line making an angle of 30° from +ve x-axis anti-clockwise. What is the force of 10 N along OP?
- (a) $(5\sqrt{3}\hat{i} + 5\hat{j})\text{N}$ (b) $(5\sqrt{3}\hat{i} - 5\hat{j})\text{N}$
 (c) $(3\sqrt{3}\hat{i} - 5\hat{j})\text{N}$ (d) $(3\sqrt{3}\hat{i} + 5\hat{j})\text{N}$
39. A particle suffers a displacement from $r_1(1,2,3)\text{m}$ to $r_2(4,3,2)\text{m}$ due to application of a force of magnitude $|\vec{F}| = 20\text{ N}$. The force acts along as vector $\vec{A} = 4\hat{j} + 3\hat{k}$. If a quantity is defined as $W = \vec{F} \cdot \Delta\vec{r}$, find W .
- (a) 20 J (b) 68 J
 (c) 4 J (d) None of these
40. The x and y component of vector $\vec{P}$ have numerical values 6 and 6, respectively, and that of $\vec{P} + \vec{Q}$ have numerical values 10 and 9. What are the numerical values of x and y components of $\vec{Q}$?
- (a) 4 and 3 respectively (b) 3 and 4, respectively
 (c) 4 and 4, respectively (d) Data insufficient
41. If $\vec{R}_1 = \vec{A} + \vec{B}$ and $\vec{R}_2 = \vec{A} - \vec{B}$, then $\frac{\vec{R}_1 + \vec{R}_2}{|\vec{R}_1 + \vec{R}_2|}$ will be along
- (a) $\vec{A}$ (b) $\vec{B}$
 (c) $\vec{A} + \vec{B}$ (d) $\vec{A} - \vec{B}$
42. The hour hand of a clock is 6 cm long. The magnitude of the displacement of the tip of hour hand between 1.00 pm to 5.00 pm is:
- (a) 6 cm (b) $6\sqrt{3}\text{ cm}$
 (c) 12 cm (d) $3\sqrt{3}\text{ cm}$
43. The velocity function of an object moving along a straight line is given by $v(t) = At^2 + Bt$. If at $t = 2\text{ s}$, the velocity is 3 m/s and acceleration is 0.500 m/s^2 , find the value of the constant A.
- (a) -0.5 ms^3 (b) -0.5 ms^2
 (c) -2.5 ms^3 (d) -1.5 ms^2
44. On the basis of given diagram, Chose the pair of vectors which have an acute angle between them:
- 
- (a) $\vec{a}$ and $\vec{c}$ (b) $\vec{a}$ and $\vec{d}$
 (c) $\vec{b}$ and $\vec{c}$ (d) $\vec{c}$ and $\vec{d}$
45. If $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} + \hat{j} - 2\hat{k}$, the angle between vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ is
- (a) 37° (b) 53°
 (c) 90° (d) 45°

RESPONSE
GRID

38. (a) (b) (c) (d) 39. (a) (b) (c) (d) 40. (a) (b) (c) (d) 41. (a) (b) (c) (d) 42. (a) (b) (c) (d)
 43. (a) (b) (c) (d) 44. (a) (b) (c) (d) 45. (a) (b) (c) (d)